

# Network Science

---

## Lecture 06: Structural Balance

5942UE Network Science

18 March 2026

Prof. Dr. Michael Granitzer Dr. Kanishika Ghosh Dastidar Dr. Jelena Mitrovic

# Structural Balance

We now study **signed networks** where relations can be positive or negative.

## Focus

- complete signed graphs
- structural balance and its global theorem
- applications to politics and online trust
- weak and relaxed balance

# Module 1

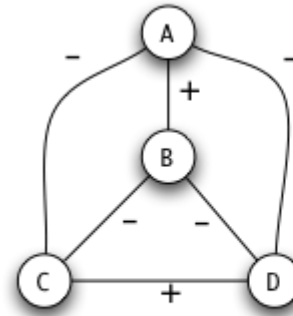
---

## Structural Balance Theorem

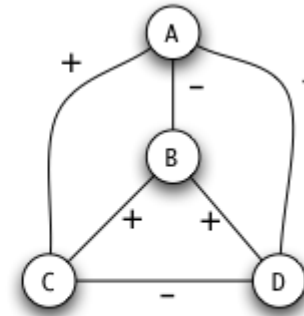
# Guiding Question

What global network structures are compatible with psychologically plausible signed triangles?

# Structural Balance Property



balanced



not balanced

A complete signed graph is **structurally balanced** if every triangle is either all-positive or has exactly one positive edge.

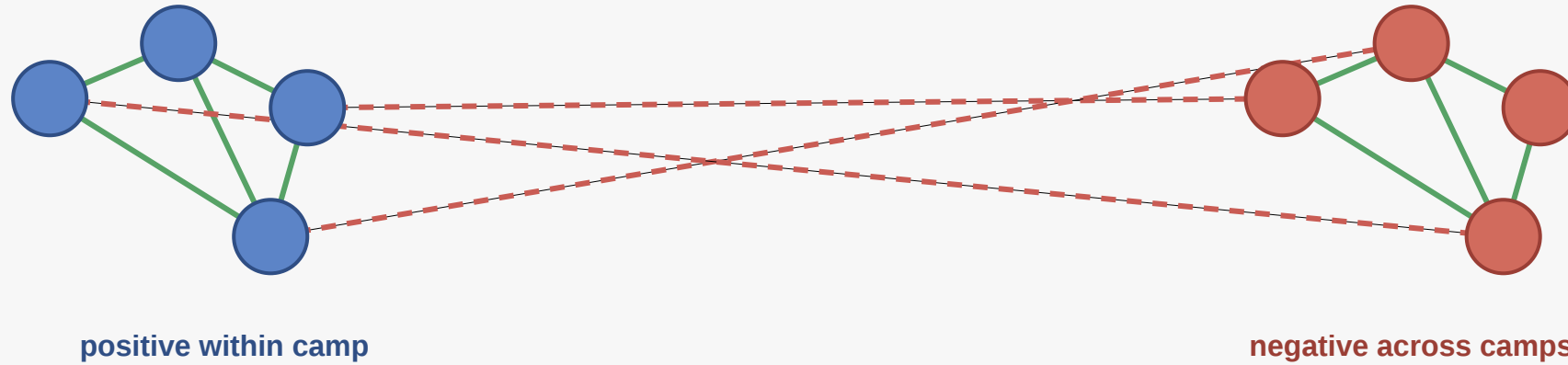
# Core Result

If a complete signed graph is balanced, then its nodes can be partitioned into:

- one all-positive block
- or two internally positive blocks with all negative edges between them

# Balanced as Camps

Balanced signed structure collapses into camps



# Interpretation

If the graph is balanced, it must collapse into either:

- one all-friends block
- or two internally friendly blocks with hostility between them

This is a classic example of a local constraint producing strong global organization.



# Module 2

---

## Applications

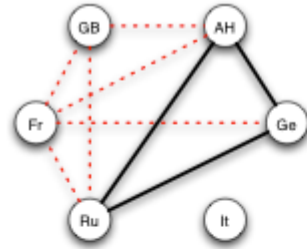
# Guiding Question

What do signed-network models help us explain in real settings?

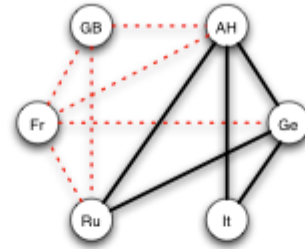
# Formal Intuition

Signed networks extend graphs by giving each edge a **positive or negative label**.

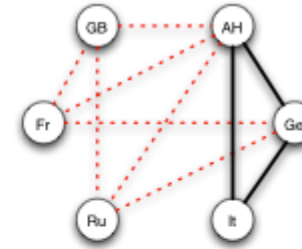
# International Relations



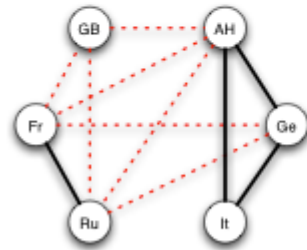
(a) *Three Emperors' League 1872-81*



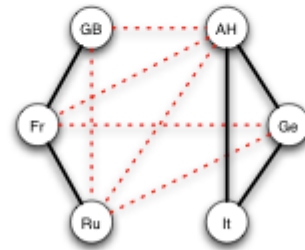
(b) *Triple Alliance 1882*



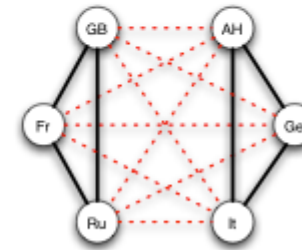
(c) *German-Russian Lapse 1890*



(d) *French-Russian Alliance 1891-94*



(e) *Entente Cordiale 1904*



(f) *British Russian Alliance 1907*

Structural balance can describe alliance formation and polarization into camps.

# Online Trust

Signed ties can also model:

- **trust and distrust**
- **recommendation and warning**
- **support and opposition**

# Interpretation

On trust platforms, positive and negative links do not behave exactly like undirected friendship and hostility, but the balance framework still suggests useful questions:

- when should distrust propagate?
- when do trust triangles stabilize?
- what patterns indicate factional structure?

Applications are strongest when the signed model is treated as an approximation, not as a literal truth

# Module 3

---

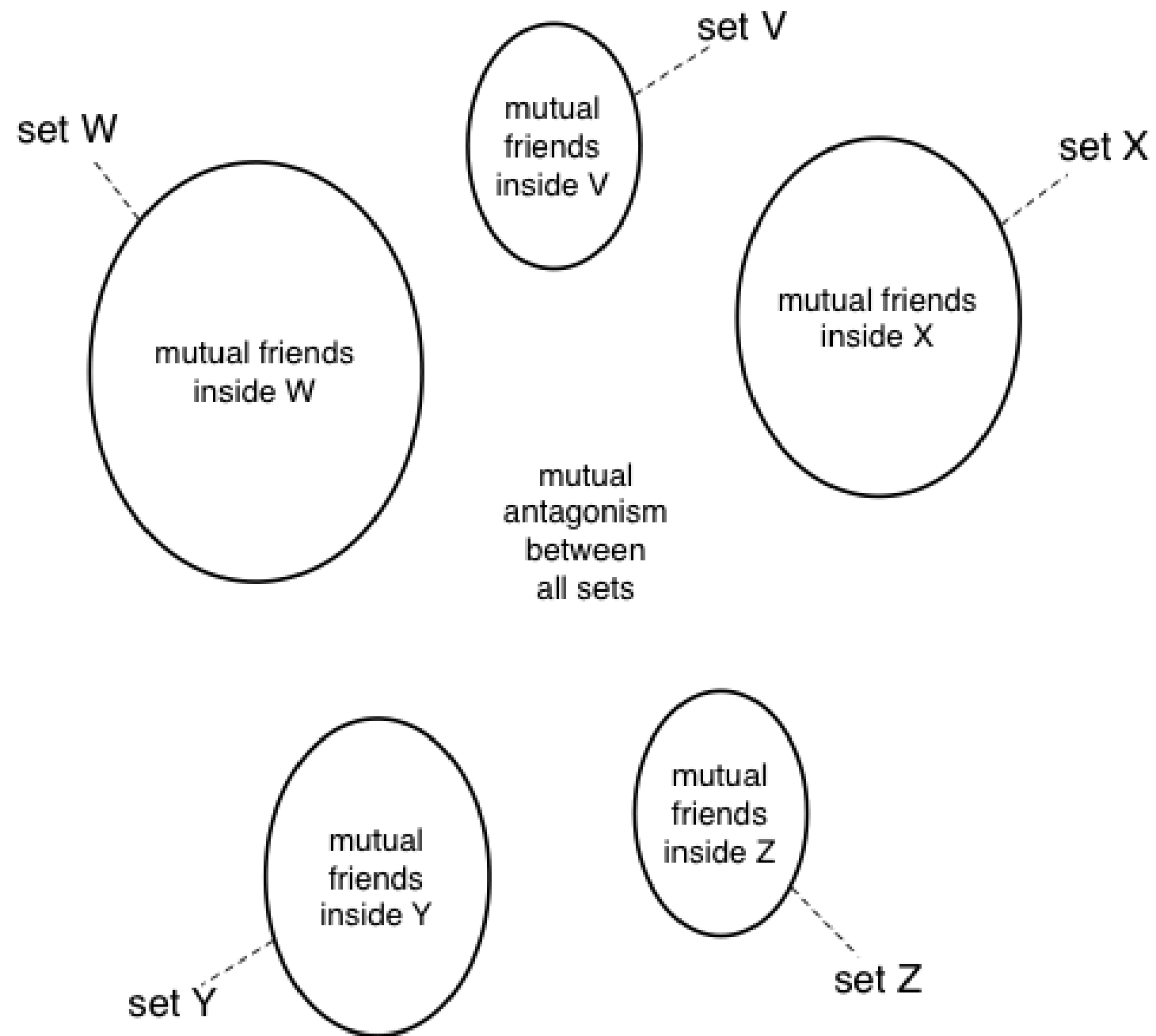
## Weak Structural Balance

# Guiding Question

What changes if "the enemy of my enemy" is not forced to become my friend?



# Counterexample



# Weak Balance

**Weak structural balance** forbids triangles with exactly two positive edges and one negative edge, but allows all-negative triangles.

# Interpretation

That relaxation changes the global theorem:

- instead of exactly two camps
- the graph may split into **multiple** internally positive groups
- with negative ties between groups

Changing one local rule can change the entire global picture.

# Module 4

---

## Relaxed and Incomplete Balance

# Guiding Question

How can we reason about balance when graphs are incomplete or only approximately balanced?

# Incomplete Graphs

Not every pair of actors has a known signed relation. That forces us to ask whether balance can still be inferred from **partial evidence**.

# Odd Negative Cycles

odd number of negative edges on a cycle  $\Rightarrow$  not balanced



# Formal Intuition

Once completeness is dropped, balance is no longer a simple triangle-only condition.

Instead, the question becomes whether the observed graph is compatible with an **approximately balanced partition**.

# Approximate Balance

In real data, a graph can be **mostly balanced** without satisfying the theorem perfectly. Then the right question is whether it is close to a balanced partition.

Relaxed balance is more realistic, but it replaces sharp theorems with softer structural diagnostics.

# Wrap-Up

- local triangle constraints imply strong global structure
- signed networks are analytically powerful but assumption-heavy
- relaxing the assumptions changes both the theorem and the interpretation